Ethan Isaac Tooms
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Education

University of Central Florida, Orlando, FL
Bachelor of Science in Computer Science | Secure Computing and Networks Minor | Mathematics Minor
Expected Graduation: Spring 2027
GPA: 3.966

Certifications

CompTIA Security+, CompTIA ITF+, Autodesk AutoCAD

Technical Skills

Languages: C, C++, Java, Python, JavaScript, HTML, CSS
Tools & Technologies: Git, Github, Bash, Wireshark, Linux, Windows, Docker, MySQL
Concepts: Data Structures, Object-Oriented Programming, Networking, Operating Systems, Cryptography, Cybersecurity Fundamentals, Machine Learning (Anomaly Detection & Classification)

Projects

Cloak: AI-Driven Network Threat Detection (Senior Design, In Progress)

- Building a network monitoring device on an Orange Pi that captures live traffic and organizes it into events for analysis
- Member of the AI team developing anomaly detection with Isolation Forest and Half-Space Trees to flag deviations from each network's learned baseline
- Applying XGBoost classification to identify known threats, including MITRE ATT&CK-mapped behavior and known malicious IPs
- Surfacing alerts in a user dashboard with a small language model (SLM) assistant to support alert investigation

Contact Manager Web Application (LAMP Stack, Team Project)

- Wrote the front-end JavaScript handling all client-side functions and API calls for a deployed contact manager with user registration and login
- Built the application's HTML/CSS pages and defined the API contract implemented by a separate back-end team
- Deployed on a Linux server (DigitalOcean) with a custom domain; source on GitHub (ethan-tooms/LAMP_Stack)

Network Packet Analysis

- Analyzed network traffic using Wireshark to inspect protocols, packet structures, and communication patterns
- Examined TCP/IP behavior, DNS queries, and HTTP/HTTPS traffic
- Applied networking and security concepts to interpret real-world data

Cryptography Implementation

- Implemented encryption/decryption algorithms in Python for (a)symmetric techniques
- Tested and validated correctness and efficiency of implemented cryptographic techniques

Concurrent Hash Table in C

- Developed a thread-safe concurrent hash table in C using POSIX threads and synchronization primitives
- Implemented fine-grained locking mechanisms to support parallel insert, delete, and lookup operations efficiently
- Optimized performance under multithreaded workloads while preventing race conditions and data corruption
- Tested scalability and correctness using concurrent stress-testing and synchronization validation techniques

C Compiler in C++

- Developed a functional compiler in C++ implementing lexical analysis, parsing, and syntax validation
- Built a custom lexer and parser to process source code and generate intermediate representations
- Applied compiler design concepts including tokenization, grammar parsing, symbol handling, and error detection
- Collaborated in debugging and testing compiler components for correctness and efficiency

MIPS Processor Simulator in C

- Collaborated in a team to design a MIPS processor simulator in C to emulate instruction execution and processor behavior
- Simulated core CPU components including registers, memory operations, instruction decoding, and ALU functionality
- Implemented support for common MIPS instructions, branching, and pipeline-related behavior
- Tested simulator accuracy using assembly programs and edge-case execution scenarios

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Experience

BJ's Wholesale Club — Recovery Clerk (2023 – Present)

- Executed store resets, reorganized inventory layouts, and built product displays during open hours
- Assigned responsibilities beyond role scope, informally recognized as "Junior PIC" by store manager
- Demonstrated strong problem-solving, adaptability, and time management in a fast-paced environment while maintaining customer interaction

Publix — Front-End Clerk (2022 – 2023)

- Supported daily store operations including customer service and closing procedures
- Collaborated with team members to ensure efficient workflow during high-traffic periods

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Leadership & Activities

KnightHawks UCF Audubon Chapter

- Participated in volunteer work building enrichment and activity structures

Knights Disc Golf Team

- Partnered with Best Buddies at UCF to teach individuals with disabilities how to play disc golf
- Participated in course cleanup and community service initiatives